A Guide to Forecast-Validation for Evaluating Conditional Volatility Forecast Models

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Abstract

This study analyzes the relationship between high-dimensional time series data through the lens of dynamic factor models, and the predictive capacity of such models in forecasting applications. Leveraging time series data from the Federal Reserve Bank of St. Louis FRED-QD database, we first identify the state space model representing this database, and then estimate the common factors using the expectation maximization algorithm with the Kalman smoother and augment the model by specifying a sparse VAR for the remaining idiosyncratic components by the means of L_1 penalized regression. Results reveal that dynamic factor models produce reliable forecasts of macroeconomic time series while simultaneously reducing the dimension of the database. Furthermore, we find that in the case of macroeconomic variables, once the common factors are extracted from the data, accounting for the remaining latent components in the database does not provide any meaningful information.

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1 Introduction

A critical task in portfolio management is forecasting the conditional variance of financial returns. Conditional variance forecasts are necessary inputs for modeling risk, pricing derivatives, and hedging portfolios. This wide-spread interest in forecasting the conditional variance has led to the rise of the enormous volatility forecasting literature, starting with the seminal paper of Engle (1982). Ever since the introduction of the ARCH and GARCH class of models (Engle (1982) and Bollerslev (1986)), a great number of models has been developed, extending the baseline ARCH framework to include more sophisticated dynamics and to incorporate more of the observed empirical regularities of the financial time-series. See A. Pagan (1996) and Poon and Granger (2003) for a comprehensive review.

This impressive pace in the development of volatility forecast models, however, had remained unmatched for the most part by model assessment methods that tend to select good models with plausible properties and satisfactory out-of-sample performance. This has led to an overwhelmingly large set of volatility forecast models that a practitioner has to choose from, which is often a difficult choice, given the degree of similarity between the models and the lack of consistent forecast selection rules. The aim of this paper is to alleviate this problem by bringing in the results of some of the most recent developments within the model selection literature and integrate them with volatility forecast models to provide an adaptable model selection procedure in a unified framework.

1.1 Relevance of Model Selection to Forecast Problems

An often challenging task during the course of an econometric analysis is the comparison of different models for modeling the particular data set at hand. Depending on the specific application, the comparison can be based on different metrics. A common metric for comparing different models, as in criterion based model selection, is the likelihood score, often penalized with the number of parameters being estimated in some way, with the penalty being higher for models with a larger number of unknown parameters. Penalizing models based on the number of parameters is rooted in the idea that by using estimates rather than the true values of the parameters, the model will drift farther apart from reality. Therefore, the information criteria generally favor more parsimonious models with fewer parameters over more complex models. The validity of the information criteria as reliable model assessment tools, however, relies heavily on the assumption that the models are not misspecified, at least not intentionally. This is a rather strong assumption that is often violated, for example in the case of Quasi-Maximum Likelihood Estimation, where the likelihood function is misspecified. See Bollerslev et al. (1994) and Sin and White (1996).

An alternative for comparing forecast models is the loss-evaluation of out-of-sample forecasts. This procedure requires the comparison of an appropriately defined loss function on the out-of-sample forecasts produced by each model, which is obtained by employing a rolling-window forecasting scheme. The choice between the rolling-window and the expanding-window forecasting schemes is often based on the assumptions surrounding the models and the data set. For models with time-varying parameters, and for series with rather short memories, the rolling-window scheme is deemed more appropriate in that it adapts to the most recent data while leaving out observations from the distant past that could become irrelevant as time moves forward within the sample. Moreover, by updating the estimates upon each rolling forward of the training sample within the data set, the time-variability of the parameter estimates can be appropriately accounted for. Once the series of forecast losses for each model is obtained, they can be compared to determine the model with the minimum overall loss.

One fundamental flaw with the direct comparison of the overall losses, however, is that it is prone to data snooping (or data mining). The effects of data snooping become prominent when multiple models are evaluated on the same data set, where some yield exceptionally smaller losses than others by chance alone, even if in reality they are equally good or even inferior. It is important, therefore, to control for the effects of data snooping in a hypothesis testing framework. The first to setup a testing framework of this kind was White (2000). The technique that White (2000) employs involves the pairwise comparison of all the models with a given benchmark, and then to determine whether the benchmark is outperformed by any of the other models in the sample in a multiple testing framework. The size of this test, which is known as the Family

Wise Error rate (FWE) is the main instrument by which tests of this kind control for data snooping.[1]

In addition to controlling for data snooping via the FWE, a desirable property for the statistical testing framework to be developed here is power. There are several different notions of power in statistical hypothesis testing. Perhaps the most familiar is the uniformly most powerful test, which has the greatest power among all tests of a given size. Even though the uniformly most powerful test can rather easily be achieved when testing for a single null hypothesis, when conducting a test of multiple hypotheses, power is restricted by the level enforced for the family wise error rate. For example, in the Bonferroni method for testing two hypotheses in a single multiple testing framework, each individual hypothesis can be conducted at most at a size $\alpha/2$ to ensure that the FWE is controlled at level α.[2] Nevertheless, some design considerations regarding the tests can lead to increased power while controlling the FWE at the same level. Two such mechanisms that we consider in this paper are the Step Wise Multiple testing procedure of Romano and Wolf (2005) and the Model Confidence Set of Peter R. Hansen et al. (2011).[3] Each of these testing mechanisms, which fall under the category of multiple comparison procedures, exploit the dependency structure of the test statistics involved to improve the power of the tests, while maintaining control over the FWE. These procedures will be discussed in detail in section 3.

A final consideration about power arises from the fact that most testing mechanisms rely on asymptotic results either for conducting the test itself or for imposing the regularity assumptions required to establish the validity of the tests. The scope of the asymptotic theory, however, is fairly limited in finite samples, and many of its results may not hold. This can significantly erode the power of the tests, unless a more powerful mechanism is implemented to accommodate for finite samples in the same way that asymptotic theory does for large samples. This can be achieved via an appropriately implemented bootstrap algorithm. As we will show later in section 3.5, the bootstrap can be used to directly estimate the density of the test statistics and to obtain the critical values for conducting the tests. We will use the bootstrap to this end, and to ultimately increase the power of our tests.

The remainder of this paper is organized as follows. Section 2 introduces the GARCH class of models and discusses their maximum likelihood estimation. Section 3 presents the validation framework and in particular the loss evaluation of forecast models. Stepwise Multiple Testing and the Model Confidence Set along with their bootstrap implementation are also covered in this section. Section 4 applies our framework to real market data and compares different volatility models for equities and FX rates. Later in the section, some diagnostic tests are presented for *ex-post* analysis of the results. Section 5 concludes and appendix A contains practical guidelines for code implementation.

2 Models of Conditional Heteroskedasticity

Early in the history of financial econometrics, it was discovered that financial return series, as statistical time-series objects exhibit certain peculiar patterns. These recorded patterns are collectively known as empirical regularities of the financial markets, and have served as the main driver of research in empirical finance. Many of the proposed models for modeling financial time-series and their dynamics are aimed at reproducing some or many of the empirical regularities that are commonly observed in the financial markets, and new discoveries on the front of empirical observation have helped refine and reassess already proposed models. One of the most prominent features concerning the empirical regularities of financial returns revolve around volatility. It is a well-established fact that financial returns exhibit volatility clustering with periods of high volatility followed by periods of high volatility and vice versa for periods of low volatility. Incidentally, in an effort to provide further refinements to the estimated variances of forecasts of macroeconomic aggregates, Engle (1982) discovers a conditionally heteroskedastic stochastic process that is capable of reproducing the volatility clustering effect that is so commonplace in financial markets. Less than four years later, in another

[1] The FWE is formally defined as the probability of incorrectly identifying at least one forecast model as superior to the benchmark (which again can happen due to the effects of data snooping).

[2] See Romano and Wolf (2005) section 2.3 and Romano and Wolf (2005) appendix D for illuminating examples.

[3] There are of course other testing procedures which have the same objectives over power and the FWE. Some examples include the SPA test of Peter Reinhard Hansen (2005), and the BRC of White (2000). Although a complete survey of all the existing methods is not possible in a single paper, the methods that we introduce in this paper are flexible and powerful enough to accommodate for most forecast selection problems that a practitioner faces on a daily basis.

ground-breaking paper, Bollerslev (1986) generalizes the ARCH model to the GARCH class (Generalized-Auto Regressive Conditional Heteroskedasticity), and lays the foundations for much of modern empirical finance, including conditional volatility forecasting in financial markets.

In the most general, abstract terms, an autoregressive conditionally heteroskedastic stochastic process $\{r_t\}_{t=1}^{T}$ is defined as:

$$r_t = h_t(\mathcal{F}_{t-1};\theta) z_t, \tag{1}$$

where $\{z_t\}_{t=1}^{T}$ is an i.i.d. process stochastically independent from the process $\{r_t\}$, and with zero conditional mean $\mathbb{E}[z_t|\mathcal{F}_{t-1}] = 0$. The function $h_t(\mathcal{F}_{t-1};\theta)$ is an $\mathcal{F}_{t-1}$-measurable function, parametrized by the parameter vector θ, with $\mathcal{F}_t$ denoting the σ-algebra induced by the process up to time t. The novelty in this specification of the return series is its non-linearity which accommodates for the correlation of non-linear transformations of the return series over time. This is plausible, since it is a very well-known fact that return series, although linearly uncorrelated, exhibit strong auto-correlation in absolute value or in quadratic terms. To see the non-linearity of the process in (1), consider the Hilbert space $\mathcal{H}_t^r$ representing the return process $\{r_t\}$ which is defined over a complete probability space $L(\Omega, \mathcal{F}_t, \mathcal{P})$, and let $\mathcal{H}_{t-1}^r \subset \mathcal{H}_t^r$ denote the Hilbert space containing the sub-process $\{r_s|s < t\}$. If we let further $\mathcal{H}_t^z$ denote the Hilbert space representing the i.i.d. process $\{z_t\}$, we can decompose $\{\mathcal{H}_t^r\}$ as:

$$\mathcal{H}_t^r = \mathcal{H}_{t-1}^r \otimes \mathcal{H}_t^z, \tag{2}$$

which defines the process $\{r_t\}$ as the tensor product between the sub-process $\{r_s|s < t\}$ and the i.i.d. process $\{z_t\}$ which is trivially non-linear.[4] The conditional variance of the return process is then given by:

$$Var\left(r_t|\mathcal{F}_{t-1}\right) = h_t^2(\mathcal{F}_{t-1};\theta), \tag{3}$$

which is heteroskedastic as it varies deterministically with time. The fundamental problem is now to specify a process for $h_t(\mathcal{F}_{t-1};\theta)$ and estimate the vector of parameters θ, which is the subject of the next sub sections. Finally, it is worth mentioning that different models differ in their specification of the function $h_t(\mathcal{F}_{t-1};\theta)$ but their modeling logic remains fundamentally the same.

2.1 Canonical GARCH Class of Models

A standard specification for the conditional variance function in (3) is the GARCH(p,q) process which is given by:

$$h_t^2 = \omega + \sum_{i=1}^{q} \alpha_i r_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i}^2. \tag{4}$$

The GARCH process is extensively studied in the literature and its properties are well known. Key features of this process are its stationarity, imposed by the constraint $\sum_{i=1}^{q} \alpha_i + \sum_{i=1}^{p} \beta_i < 1$ and its symmetry, implying that both positive and negative returns influence future volatility equally. The baseline GARCH process has evolved over time and much of this evolution has been guided by empirical findings. A prominent finding that dates back to very early in the history of the literature is that negative returns have a larger impact on future volatility than positive returns of the same magnitude. This effect, commonly referred to as the leverage effect was first discovered by Black (1976) and later illustrated by A. R. Pagan and Schwert (1990) and Engle and Ng (1993) in the context of the news impact curve. A natural way to incorporate the leverage effect into the GARCH process is to adjust the coefficients of the lagged squared returns so that the coefficients are smaller when the returns are positive. This is the GJR-GARCH process of Glosten et al. (1993):

$$h_t^2 = \omega + \sum_{i=1}^{q} \left(\alpha_i + \gamma_i \mathbb{I}_{[r_{t-i}>0]}\right) r_{t-i}^2 + \sum_{i=1}^{p} h_{t-i}^2. \tag{5}$$

[4] It is worth mentioning that linear processes can always be decomposed as the direct sum of two or more orthogonal processes of the form: $\mathcal{H}_t^r = \mathcal{H}_{t-1}^r \oplus \mathcal{H}_{t-1}^{r\perp}$ where $\mathcal{H}_{t-1}^{r\perp}$ is the orthogonal complement of $\mathcal{H}_{t-1}^r$.

An alternative approach is to model the natural logarithm of the variance as in the EGARCH model of Nelson (1991):

$$log(h_t^2) = \omega + \sum_{i=1}^{q} \left[\alpha_i z_{t-i} + \gamma_i \left(|z_{t-i}| - \mathbb{E}\left[z_{t-i}\right]\right)\right] + \sum_{i=1}^{p} \beta_i h_{t-i}^2, \tag{6}$$

where $z_{t-i} = r_{t-i}/h_{t-i}$ is the standardized return. The leverage effect in the EGARCH process is manifested by the coefficient $\gamma_i < 0$. Another significant discovery was that absolute returns have a stronger serial correlation than squared returns. This was formally shown by Ding et al. (1993), and led to the proposal of the Asymmetric Power ARCH (APARCH) model:

$$h_t^\delta = \omega + \sum_{i=1}^{q} \alpha_i \left[|r_{t-i} - \gamma_i r_{t-i}|\right]^\delta + \sum_{i=1}^{p} \beta_i h_{t-i}^\delta. \tag{7}$$

By allowing the degree the degree of the process δ to be a free parameter, the APARCH model is more flexible in capturing the dynamics of the conditional variance than other specifications. There exist many other models for the conditional variance, and the list of the models presented here is by no means exhaustive. Nevertheless, the models mentioned so far are among the most commonly used and nest many of the other specifications. See Ding et al. (1993) and Hentschel (1995).

2.2 Maximum Likelihood Estimation

The parameters of the GARCH-type processes[5] are naturally estimated with the maximum likelihood estimator. Under Fisher regularity conditions, the maximum likelihood estimator achieves the Cramer-Rao bound, and is therefore asymptotically efficient. An exception, however, may arise if the Fisher information matrix is singular, which happens if the fourth-order moment of the underlying process is infinite. In such cases, the (quasi)-maximum likelihood estimator may still be consistent, however, it will no longer be efficient. Under these circumstances, practitioners may prefer to use a heavy-tailed distribution for the likelihood function such as the Student-t. It must be noted, however, that this can introduce bias into the estimator. In fact, Hall and Yao (2003) show that within the class of maximum likelihood estimators, the Gaussian QMLE is the only distribution-free, and therefore consistent estimator. Any other misspecified likelihood function is bound to be asymptotically biased. Therefore, any possible gains in efficiency must be weighed against the potential bias due to the misspecification of the likelihood function.

Maximum likelihood estimation of the GARCH-type processes with the Gaussian likelihood function is a standard practice and can be found in most modern text books and related articles. See Hamilton (1994), Engle (1982), and Bollerslev (1986) among others. An important contribution is due to Hall and Yao (2003) where they show that the asymptotic distribution of the Gaussian QMLE is Levy-stable (although not Gaussian) and that its rate of convergence is $T^{1-\alpha/2}$, where α is the tail index of the density of innovations in the process, z_t. Moreover, they show that the (robust) asymptotic covariance matrix of the estimator is given by:

$$\Sigma = H^{-1} V H^{-1}, \tag{8}$$

where V is the asymptotic covariance matrix of the score function and H^-1 is the inverse of the Hessian of the likelihood function.

Regardless of the choice of the estimator for GARCH-type processes, the conditions for the existence of the fourth-order moment of the underlying processes can be driven analytically. These conditions, however, are generally unknown and must be driven for each specific process individually.[6] Therefore, in practical applications, it is more convenient to employ an empirical method for testing for the finiteness of the fourth-order moment of the process *ex-post*, such as estimating the tail index of the standardized residuals. See section 4.2 for more details.

[5]We refer to the GARCH class of processes which include the GARCH process and its different varieties collectively as the GARCH-type processes.

[6]The necessary and sufficient condition for the GARCH(1,1) process for instance is: $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$.

3 Validation Framework

3.1 Rolling Window Estimation Scheme

A central difficulty in volatility forecasting is that the parameters of financial time-series models are rarely stable over long periods. Financial markets are frequently subject to structural breaks, changing persistence patterns, and volatility regime shifts. Consequently, parameter estimates obtained from distant historical observations may become uninformative or even detrimental for future forecasts. This issue is particularly important in risk management applications, where forecast accuracy depends critically on the model's ability to adapt to new market conditions. To address this problem, we employ a rolling window estimation scheme.

Let $\{r_t\}_{t=1}^{T}$ denote a sequence of demeaned returns and let W denote the size of the estimation window. For a given forecast model indexed by $m \in \mathcal{M}$, the model is estimated repeatedly on rolling subsamples of fixed length W. At each iteration t, the model is estimated using the observations:

$$\{r_{t-W+1}, r_{t-W+2}, \dots, r_t\}, \tag{9}$$

and a one-step-ahead forecast of the conditional variance is produced:

$$\hat{h}^2_{t+1|t,m}. \tag{10}$$

The window is then shifted forward by one observation and the procedure is repeated recursively until the end of the sample is reached. Therefore, for each competing model, a sequence of out-of-sample volatility forecasts is generated:

$$\left\{\hat{h}^2_{W+1|W,m}, \hat{h}^2_{W+2|W+1,m}, \dots, \hat{h}^2_{T|T-1,m}\right\}. \tag{11}$$

A further justification for the rolling estimation scheme arises from the asymptotic theory of forecast comparison procedures. Let

$$d_{ij,t} = L_{i,t} - L_{j,t}, \tag{12}$$

denote the loss differential between two competing forecast models i and j, where $L_{i,t}$ and $L_{j,t}$ are the corresponding forecast losses at time t. Forecast comparison procedures such as the Superior Predictive Ability and the Model Confidence Set rely on the assumption that the sequence of loss differentials satisfies suitable stationarity and weak dependence conditions.

In practice, recursive parameter estimation on expanding samples may lead to substantial parameter uncertainty and unstable forecast loss dynamics, particularly in the presence of structural breaks. Rolling window estimation alleviates some of these issues by stabilizing the distribution of the forecast loss differentials and reducing the influence of distant observations. As emphasized by Giacomini and White (2006) and Peter R. Hansen et al. (2011), the estimation scheme itself forms an integral part of the forecast evaluation framework and must therefore be explicitly accounted for in predictive ability analysis.

3.2 Loss-Evaluation of Forecast Models

—— XXX We evaluate and compare different forecast models by their average out-of-sample forecast losses. To this end, we would need to compute the loss entailed by each model by comparing the model forecasts with the observed volatilities. An immediate problem that arises, however, is that volatility is a latent variable and it cannot be observed. Therefore, we have to rely on (imperfect) proxies for the loss-evaluation of the forecast models. Further exacerbating the problem is the fact that loss functions are generally not robust to the noise involved in the proxies. This can lead to inconsistent evaluation schemes with often contradictory results, making the entire loss-evaluation procedure inefficient and even invalid. See A. R. Pagan and Schwert (1990), Bollerslev and Ghysels (1996) and Andersen et al. (1999) for a discourse on this issue.

Instead of "real" use the term "latent" to refer to the unobserved volatilities.

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Another fundamental problem to address in forecasting variables is the inability to observe the variable of interest, even *ex-post*. This issue is of crucial importance in volatility forecasting, which is the main subject treated in this paper, and can be resolved at least partially, since unbiased estimators are available. Two of the most used conditionally unbiased proxies are squared returns and realized volatility.

Comparisons between volatility forecasts are based on the distance, or expected loss, to the true conditional variance. Patton (2011) focuses on the definition of the loss function since, if the ranking of two forecasts obtained from a so-called *imperfect* volatility proxy is the same as the ranking that would be obtained from the real variable, which is actually unfeasible, then the loss function can be identified as "robust". A robust loss function allows for the comparison of the accuracy of a forecast, on average, even if the real variable of interest is not observable.

However, a robust loss function does not automatically make the proxy choice irrelevant. Its role is to guarantee that the comparison between forecasts is not systematically distorted by the usage of an imperfect variable in place of the real conditional variance. Nevertheless, the proxy still influences the noise of the valuation, the precision of the loss differential estimations, and in the end, the following statistical inference. This distinction is particularly important in the comparison between squared returns and realized variance. The first are easy to compute and available for every series of daily prices, but are a very noisy proxy of the conditional variance. Realized variance, on the other hand, built from intraday returns, gives a more informative measure of the effectively realized volatility, still introducing practical problems such as microstructure noise and data availability, given the intraday nature of the series used.

Formally, let $\hat{\sigma}_t^2$ be the real conditional variance, and let $\hat{\sigma}_t^2$ be a proxy observable ex-post, such as squared returns or realized variance. A loss function $L(\cdot,\cdot)$ can be defined as robust if the expected ranking between the two forecasts is preserved when the real conditional variance is substituted by its observable proxy. Following Patton (2011), if model i produces a better forecast than model j, with respect to the real conditional variance, so if

$$E\left[L(\hat{\sigma}_t^2, \hat{h}_{i,t}^2)\right] < E\left[L(\hat{\sigma}_t^2, \hat{h}_{j,t}^2)\right],$$

then a robust loss function should preserve the same order when the evaluation is done using the proxy

$$E\left[L(\hat{\sigma}_t^2, \hat{h}_{i,t}^2)\right] < E\left[L(\hat{\sigma}_t^2, \hat{h}_{j,t}^2)\right].$$

Going further in the research, two main robust loss functions are considered: Mean Squared Error (MSE) and Quasi-Likelihood loss (QLIKE). Both are proven to be robust in the sense already discussed. In Patton (2011), this proof is obtained under three different assumptions for the conditional distribution of daily returns:

$$r_t \,|\mathcal{F}_{t-1} \sim \begin{cases} F_t(0, \sigma_t^2) \\ \text{Student's } t(0, \sigma_t^2, \nu) \\ \mathcal{N}(0, \sigma_t^2) \end{cases}$$

where $F_t(0, \sigma_t^2)$ is some unspecified distribution with null mean and variance σ_t^2, and Student's $t(0, \sigma_t^2, \nu)$ is a Student's t distribution with mean zero, variance σ_t^2, and ν degrees of freedom. In all these cases, the squared daily returns, used for the estimation, are a valid volatility proxy. Amongst all nine widely used loss functions analyzed in the already cited paper, only MSE and QLIKE lead to an optimal forecast $h_t^* = \mathbb{E}_{t-1}[\hat{\sigma}_t] = \sigma_t$ and so can be considered robust.

Mean Squared Error is defined as:

$$L_{MSE}(\hat{\sigma}_t^2, \hat{h}_{i,t}^2) = (\hat{\sigma}_t^2 - \hat{h}_{i,t}^2)^2.$$

MSE measures the quadratic distance between the realized variance proxy and the model forecast. His main upside is the fact that it is immediately interpreted: wider forecasting errors are increasingly penalized, yielding an easy-to-compute and report metric. On the other hand, it is very sensitive to outliers, since a few extreme volatility episodes can have a very high impact on the average loss.

Quasi-Likelihood, instead, is defined as

$$L_{QLIKE}(\hat{\sigma}_t^2, \hat{h}_{i,t}^2) = \log(\hat{h}_{i,t}^2) + \frac{\hat{\sigma}_t^2}{\hat{h}_{i,t}^2}$$

QLIKE has a natural quasi-likelihood interpretation, and it is particularly relevant since it penalizes significantly forecasts that underestimate realized volatility, specifically during high turmoil periods.

The joint application of MSE and QLIKE allows evaluating models performance from complementary perspectives. MSE emphasizes absolute forecast error and is easy to interpret, while QLIKE is more linked to the logic of conditional variance models. If a model is competitive under both metrics, then the obtained ranking is more robust. If, instead, the results differ, this difference is still informative since it signals that the relative performance depends on the adopted accuracy definition.

Later on in this paper, the main reference for the evaluation of the different models' performance will be these two loss functions.

Operatively, once the loss function is chosen, the comparison between two models i and j is performed through the loss differential:

$$d_{ij,t} = L(\hat{\sigma}_t^2, \hat{h}_{i,t}^2) - L(\hat{\sigma}_t^2, \hat{h}_{j,t}^2).$$

Hence, the loss differential series represents the central empirical object on which the model comparison procedures are based. Following Peter R. Hansen et al. (2011), we assume that the process $\{d_{ij,t}\}$ is strictly stationary with finite moments and satisfies an α-mixing condition. Importantly, these assumptions are imposed on the loss differentials rather than directly on the individual loss processes themselves. Consequently, certain forms of structural instability and nonstationarity may still be admissible, provided that they affect competing models in a sufficiently similar manner so that the stationarity of the loss differential process is preserved.

3.3 Stepwise Multiple Testing

In empirical forecast evaluation, competing models are often compared across a large number of alternative specifications. In such settings, superior forecasting performance may arise purely due to random variation rather than genuine predictive ability. This issue is commonly referred to as data snooping or data mining and represents one of the central challenges in empirical forecast evaluation.

Suppose that a collection of competing volatility forecast models

$$\mathcal{M} = \{M_1, M_2, \ldots, M_m\} \tag{13}$$

is evaluated on the basis of their out-of-sample forecast losses. Even if all competing models possess identical predictive ability, repeated comparisons across many specifications increase the probability of falsely identifying at least one model as superior. Consequently, conventional pairwise testing procedures may produce misleading inference when applied repeatedly over a large universe of competing models.

The data snooping problem is particularly severe in empirical finance, where forecasting models are frequently modified, re-estimated, and compared across alternative specifications. As emphasized by Romano and Wolf (2005), repeated hypothesis testing without appropriate correction can substantially inflate the probability of false discoveries and lead to overoptimistic conclusions regarding predictive performance.

To address this issue, Romano and Wolf (2005) propose a stepwise multiple testing procedure that evaluates the predictive performance of the entire collection of models simultaneously while asymptotically controlling the familywise error rate.

To formalize the multiple testing problem, let

$$H_s : \theta_s \leq 0, \qquad H_s' : \theta_s > 0, \tag{14}$$

for $s = 1, \ldots, S$, denote the collection of hypotheses associated with the competing forecasting models. Here, θ_s represents the predictive performance measure of model s relative to a benchmark or reference model.

The objective is to identify as many models as possible for which $\theta_s > 0$ while controlling the familywise error rate asymptotically.

Let

$$w_{T,s} \tag{15}$$

denote the test statistic associated with hypothesis H_s. The stepwise procedure begins by ordering the statistics from largest to smallest,

$$w_{T,r_1} \geq w_{T,r_2} \geq \cdots \geq w_{T,r_S}, \tag{16}$$

where the relabeling indices $(r_1, \ldots, r_S)$ correspond to the ordered hypotheses.

The central idea of Romano and Wolf (2005) is to construct joint confidence regions for the vector

$$(\theta_{r_1}, \ldots, \theta_{r_S})', \tag{17}$$

with nominal joint coverage probability $1 - \alpha$. In the first step, the joint confidence region takes the rectangular form

$$[w_{T,r_1} - c_1, \infty) \times \cdots \times [w_{T,r_S} - c_1, \infty), \tag{18}$$

where the common critical value c_1 is chosen such that the confidence region attains the required asymptotic coverage probability.

The hypotheses are then tested within this confidence region. Specifically, if the interval

$$[w_{T,r_s} - c_1, \infty) \tag{19}$$

does not contain zero, the corresponding null hypothesis H_{r_s} is rejected.

Unlike conventional single-step procedures, however, the stepwise methodology continues after the first rejection stage. Once rejected hypotheses are removed, a new joint confidence region is constructed over the reduced hypothesis space,

$$[w_{T,r_{R_1+1}} - c_2, \infty) \times \cdots \times [w_{T,r_S} - c_2, \infty), \tag{20}$$

where the updated critical value c_2 is recalibrated over the remaining hypotheses. The procedure is then repeated sequentially until no additional hypotheses can be rejected.

The recursive updating of the confidence regions constitutes the primary source of the power improvement of the procedure. Standard single-step methods construct critical values that protect against the worst-case multiple testing scenario over the entire hypothesis space and therefore tend to be excessively conservative. In contrast, the stepwise procedure progressively reduces the dimension of the hypothesis space and recomputes the associated confidence regions at each stage. This leads to less conservative critical values and substantially improves the probability of rejecting false null hypotheses while still asymptotically controlling the familywise error rate.

In practice, the critical values $c_1, c_2, \ldots$ depend on the unknown sampling distribution of the statistics and are therefore approximated through bootstrap resampling methods. By estimating the joint dependence structure of the statistics directly from the data, the bootstrap procedure further improves the power of the tests relative to traditional correction methods such as Bonferroni-type procedures.

The exposition above follows the basic StepM framework of Romano and Wolf (2005) based on generic test statistics $w_{T,s}$. In empirical applications, however, studentized statistics are often preferable because the variability of the test statistics may differ substantially across competing models. The studentized statistic is defined as

$$z_{T,s} = \frac{w_{T,s}}{\hat{\sigma}_{T,s}}, \tag{21}$$

where $\hat{\sigma}_{T,s}$ denotes an estimator of the standard deviation of $w_{T,s}$. The hypotheses are then ordered according to the studentized statistics rather than the raw statistics:

$$z_{T,r_1} \geq z_{T,r_2} \geq \cdots \geq z_{T,r_S}. \tag{22}$$

The corresponding bootstrap critical value at step j is obtained from the distribution of the maximum studentized statistic over the remaining hypotheses:

$$\hat{d}_j = \inf \left\{ x : \mathbb{P}_{\hat{P}_T} \left(\max_{R_{j-1}+1 \leq s \leq S} \frac{w^*_{T,r_s} - \theta^*_{T,r_s}}{\hat{\sigma}^*_{T,r_s}} \leq x \right) \geq 1 - \alpha \right\}. \tag{23}$$

The null hypothesis H_{r_s} is rejected whenever the corresponding studentized confidence interval excludes zero:

$$0 \notin \left[w_{T,r_s} - \hat{\sigma}_{T,r_s} \hat{d}_j, \infty \right). \tag{24}$$

Thus, studentization changes the scale on which the hypotheses are ordered and tested, but it does not change the logic of the StepM procedure. The method still constructs joint confidence regions, eliminates rejected hypotheses, and recomputes the critical values over the reduced hypothesis space. Its main advantage is that it accounts for heteroskedasticity and scale differences across test statistics, which makes the comparisons more balanced in finite samples.

The Model Confidence Set procedure developed by Peter R. Hansen et al. (2011), presented in the following section, extends this sequential testing logic to identify the subset of models that can be regarded as statistically indistinguishable from the best model at a given confidence level.

3.4 Model Confidence Set

———- XXX Mention MCS p-values and the mechanics of the elimination rule.

———- XXX

A simple ranking of models competing with respect to the average loss is not sufficient to determine which one is the best model. Even if one of them presents the lowest sample loss, the difference observed between the remaining alternatives could be too small to be statistically significant. As Peter R. Hansen et al. (2011) highlights, in many empirical applications, data do not contain sufficient information to univocally identify a single dominant model. In these cases, forcing the selection of a unique winner is likely to produce excessively confident conclusions compared to the evidence that is effectively available.

The Model Confidence Set (MCS) procedure, introduced by Peter R. Hansen et al. (2011), deals with this problem by substituting the search for a single best model with the construction of a set of models that can include the best one or ones, at a fixed confidence level. The authors describe the MCS as the equivalent, within the problem of model selection, of a confidence interval for a parameter: when data are very informative, the final set can include few models and even reduce to just one; when, instead, data are less informative, the set remains broader, reflecting the inability to cleanly distinguish between the competing alternatives.(Peter R. Hansen et al. 2011)

Let $\mathcal{M}^0$ be the initial set of candidate models and, as seen in the previous sections, let

$$d_{ij,t} = L_{i,t} - L_{j,t}$$

the loss differential between models i and j at time t. Hence, the expected relative performance between two models is defined as

$$\mu_{ij} = E[d_{ij,t}].$$

If $\mu_{ij} < 0$, model i presents a lower expected loss than model j and results preferable with respect to the considered loss function. From this quantity, Peter R. Hansen et al. (2011) define the superior models set as

$$\mathcal{M}^* = \left\{ i \in \mathcal{M}^0 : \mu_{ij} \leq 0 \qquad \forall j \in \mathcal{M}^0 \right\}.$$

Hence, the set $\mathcal{M}^*$ contains all models whose expected loss is no greater than that of any other model in the initial set. It is important to observe that a set can include more than one element: the procedure does not assume, nor requires, that a unique best model exists.

The aim of the MCS is the estimation of $\mathcal{M}^*$ through a sequential procedure based on equivalence testing and an elimination rule. For each current subset $\mathcal{M} \subseteq \mathcal{M}^0$, the null hypothesis of equal predictive ability is tested:

$$H_{0,\mathcal{M}} : \mu_{ij} = 0 \quad \text{per ogni } i, j \in \mathcal{M}.$$

This hypothesis states that, within the current set, no model has expected predictive performance that differs from that of the others. If the null hypothesis is not rejected the procedure stops, and the current set can be identified as the Model Confidence Set. If, instead, it gets rejected, there is evidence that at least one of the models within $\mathcal{M}$ is inferior to the others. One elimination rule removes the model with the worst relative performance, and the test is repeated on the reduced subset. This iterative elimination algorithm constitutes the core of the procedure proposed by Peter R. Hansen et al. (2011).

The final result is a random set of models, denoted by

$$\widehat{\mathcal{M}}^*_{1-\alpha},$$

built in such a way to contain the real superior models subset with probability at least equal to $1 - \alpha$ in an asymptotic sense:

$$\liminf_{n\to\infty} P\left(\mathcal{M}^* \subseteq \widehat{\mathcal{M}}^*_{1-\alpha}\right) \geq 1 - \alpha.$$

This property justifies the term *confidence set*: MCS does not claim to always identify a unique winner, but is constructed so that the truly superior models are excluded only with a controlled probability. In finite samples, the final set can still contain inferior models. As the authors clarify, in fact, a model is removed only when it results significantly worse than at least one alternative, and not all models that survive the procedure are necessarily to be interpreted as good models in an absolute sense.

In this research context, the MCS is applied to the loss differential series generated by MSE and QLIKE loss functions. In this way, the procedure makes it possible to identify, for each evaluation criterion, the set of volatility models that cannot be considered significantly inferior to the best models within the initial set. The MCS addresses a different question compared to stepwise multiple testing discussed in the previous section. While the latter verifies which models outperform a pre-specified benchmark, MCS determines which models belong to the best models set without imposing an ex-ante reference model, feature that Peter R. Hansen et al. (2011) identify as one of the principal benefits of this procedure compared to the superior predictive ability tests.

3.5 Bootstrap Implementation

Unless the sample size is large enough relative to the number of models being considered for the test statistics to uniformly converge to their asymptotic distributions, it is not possible to obtain precise estimates of the covariance matrix of the test statistics for carrying out the tests. Furthermore, the test statistics considered here have non-standard asymptotic distributions since they depend on nuisance parameters. Therefore, it is convenient to use a bootstrap implementation that that does not require an explicit estimate of the covariance matrix and that it can handle the issue related to nuisance parameters implicitly. By appropriately employing a bootstrapping algorithm, we can directly compute the critical values for constructing the confidence regions in the StepM procedure and directly estimate the MCS p-values.

4 Empirical Analysis

4.1 Results

4.2 Diagnostic Tests

5 Conclusion

A Code Implementation

References

Andersen, Torben G, Tim Bollerslev, and Steve Lange (1999). “Forecasting financial market volatility: Sample frequency vis-à-vis forecast horizon”. In: *Journal of Empirical Finance* 6.5, pp. 457–477. ISSN: 0927-5398. DOI: https://doi.org/10.1016/S0927-5398(99)00013-4. URL: https://www.sciencedirect.com/science/article/pii/S0927539899000134.

Andrews, Donald W.K. (1991). “Asymptotic optimality of generalized CL, cross-validation, and generalized cross-validation in regression with heteroskedastic errors”. In: *Journal of Econometrics* 47.2, pp. 359–377. ISSN: 0304-4076. DOI: https://doi.org/10.1016/0304-4076(91)90107-O. URL: https://www.sciencedirect.com/science/article/pii/030440769190107O.

Bergmeir, Christoph, Rob J. Hyndman, and Bonsoo Koo (2018). “A note on the validity of cross-validation for evaluating autoregressive time series prediction”. In: *Computational Statistics Data Analysis* 120, pp. 70–83. ISSN: 0167-9473. DOI: https://doi.org/10.1016/j.csda.2017.11.003. URL: https://www.sciencedirect.com/science/article/pii/S0167947317302384.

Black, Fisher (1976). “Studies of Stock Price Volatility Changes”. In: *Proceedings of the 1976 Business and Economic Statistics Section*. American Statistical Association, pp. 177–181.

Bollerslev, Tim (1986). “Generalized autoregressive conditional heteroskedasticity”. In: *Journal of Econometrics* 31.3, pp. 307–327. ISSN: 0304-4076. DOI: https://doi.org/10.1016/0304-4076(86)90063-1. URL: https://www.sciencedirect.com/science/article/pii/0304407686900631.

Bollerslev, Tim, Robert F. Engle, and Daniel B. Nelson (1994). “Chapter 49 Arch models”. In: vol. 4. Handbook of Econometrics. Elsevier, pp. 2959–3038. DOI: https://doi.org/10.1016/S1573-4412(05)80018-2. URL: https://www.sciencedirect.com/science/article/pii/S1573441205800182.

Bollerslev, Tim and Eric Ghysels (1996). “Periodic Autoregressive Conditional Heteroscedasticity”. In: *Journal of Business Economic Statistics* 14.2, pp. 139–151. ISSN: 07350015. URL: http://www.jstor.org/stable/1392425 (visited on 03/17/2026).

Brownlees, Christian, Robert Engle, and Bryan Kelly (2012). “A practical guide to volatility forecasting through calm and storm”. In: *Journal of Risk* 14.2, pp. 3–22. DOI: https://doi.org/10.21314/JOR.2012.237.

Ding, Zhuanxin, Clive W.J. Granger, and Robert F. Engle (1993). “A long memory property of stock market returns and a new model”. In: *Journal of Empirical Finance* 1.1, pp. 83–106. ISSN: 0927-5398. DOI: https://doi.org/10.1016/0927-5398(93)90006-D. URL: https://www.sciencedirect.com/science/article/pii/092753989390006D.

Engle, Robert F. (1982). “Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation”. In: *Econometrica* 50.4, pp. 987–1007. ISSN: 00129682, 14680262. URL: http://www.jstor.org/stable/1912773 (visited on 03/17/2026).

Engle, Robert F. and Victor K. Ng (1993). “Measuring and Testing the Impact of News on Volatility”. In: *The Journal of Finance* 48.5, pp. 1749–1778. DOI: https://doi.org/10.1111/j.1540-6261.1993.tb05127.x. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.1993.tb05127.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.1993.tb05127.x.

Giacomini, Raffaella and Halbert White (2006). “Tests of Conditional Predictive Ability”. In: *Econometrica* 74.6, pp. 1545–1578.

Glosten, Lawrence R., Ravi Jagannathan, and David E. Runkle (1993). “On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks”. In: *The Journal of Finance* 48.5, pp. 1779–1801. DOI: https://doi.org/10.1111/j.1540-6261.1993.tb05128.x. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.1993.tb05128.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.1993.tb05128.x.

Hall, Peter and Qiwei Yao (2003). “Inference in Arch and Garch Models with Heavy–Tailed Errors”. In: *Econometrica* 71.1, pp. 285–317. DOI: https://doi.org/10.1111/1468-0262.00396. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1468-0262.00396. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0262.00396.

Hamilton, J.D. (1994). *Time Series Analysis*. Princeton University Press, Princeton. DOI: https://doi.org/10.1515/9780691218632.

Hansen, Peter R. and Asger Lunde (2005). “A forecast comparison of volatility models: does anything beat a GARCH(1,1)?” In: *Journal of Applied Econometrics* 20.7, pp. 873–889. DOI: https://doi.org/10.

1002/jae.800. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/jae.800. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/jae.800.

Hansen, Peter R., Asger Lunde, and James M. Nason (2011). "The Model Confidence Set". In: *Econometrica* 79.2, pp. 453–497.

Hansen, Peter Reinhard (2005). "A Test for Superior Predictive Ability". In: *Journal of Business Economic Statistics* 23.4, pp. 365–380. ISSN: 07350015. URL: http://www.jstor.org/stable/27638834 (visited on 03/25/2026).

Hastie, Trevor et al. (2004). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction.* DOI: https://doi.org/10.1007/BF02985802.

Hentschel, Ludger (1995). "All in the family Nesting symmetric and asymmetric GARCH models". In: *Journal of Financial Economics* 39.1, pp. 71–104. ISSN: 0304-405X. DOI: https://doi.org/10.1016/0304-405X(94)00821-H. URL: https://www.sciencedirect.com/science/article/pii/0304405X9400821H.

Li, Ker-Chau (1987). "Asymptotic Optimality for Cp, CL, Cross-Validation and Generalized Cross-Validation: Discrete Index Set". In: *The Annals of Statistics* 15.3, pp. 958–975. ISSN: 00905364, 21688966. URL: http://www.jstor.org/stable/2241809 (visited on 03/18/2026).

Majidi, Shahriyar, Pietro Sgrigna, and Tommaso Zeri (2026). *Forecast-Validation for optimal ARCH/GARCH model selection in Python.* Tech. rep. Starting Finance Club. URL: https://github.com/donn-majidi/sfclubunibo-arch-fv.

Nelson, Daniel B. (1991). "Conditional Heteroskedasticity in Asset Returns: A New Approach". In: *Econometrica* 59.2, pp. 347–370. ISSN: 00129682, 14680262. URL: http://www.jstor.org/stable/2938260 (visited on 03/25/2026).

Pagan, Adrian (1996). "The econometrics of financial markets". In: *Journal of Empirical Finance* 3.1, pp. 15–102. ISSN: 0927-5398. DOI: https://doi.org/10.1016/0927-5398(95)00020-8. URL: https://www.sciencedirect.com/science/article/pii/0927539895000208.

Pagan, Adrian R. and G. William Schwert (1990). "Alternative models for conditional stock volatility". In: *Journal of Econometrics* 45.1, pp. 267–290. ISSN: 0304-4076. DOI: https://doi.org/10.1016/0304-4076(90)90101-X. URL: https://www.sciencedirect.com/science/article/pii/030440769090101X.

Patton, Andrew J. (2011). "Volatility forecast comparison using imperfect volatility proxies". In: *Journal of Econometrics* 160.1. Realized Volatility, pp. 246–256. ISSN: 0304-4076. DOI: https://doi.org/10.1016/j.jeconom.2010.03.034. URL: https://www.sciencedirect.com/science/article/pii/S030440761000076X.

Poon, Ser-Huang and Clive W.J. Granger (2003). "Forecasting Volatility in Financial Markets: A Review". In: *Journal of Economic Literature* 41.2, pp. 478–539. DOI: 10.1257/002205103765762743. URL: https://www.aeaweb.org/articles?id=10.1257/002205103765762743.

Romano, Joseph P. and Michael Wolf (2005). "Stepwise Multiple Testing as Formalized Data Snooping". In: *Econometrica* 73.4, pp. 1237–1282. DOI: 10.1111/j.1468-0262.2005.00615.x.

Sin, Chor-Yiu and Halbert White (1996). "Information criteria for selecting possibly misspecified parametric models". In: *Journal of Econometrics* 71.1, pp. 207–225. ISSN: 0304-4076. DOI: https://doi.org/10.1016/0304-4076(94)01701-8. URL: https://www.sciencedirect.com/science/article/pii/0304407694017018.

West, Kenneth D. (1996). "Asymptotic Inference about Predictive Ability". In: *Econometrica* 64.5, pp. 1067–1084. ISSN: 00129682, 14680262. URL: http://www.jstor.org/stable/2171956 (visited on 03/18/2026).

White, Halbert (2000). "A Reality Check for Data Snooping". In: *Econometrica* 68.5, pp. 1097–1126. ISSN: 00129682, 14680262. URL: http://www.jstor.org/stable/2999444 (visited on 05/11/2026).